

Operations & Maintenance Procedures (Excerpt)

SAMPLE / SYNTHETIC DOCUMENT - Section 4: Inverter and BESS Maintenance. Prepared for demonstration purposes only.

4.1 Routine Inspection Schedule

Routine inspections of inverters and battery energy storage system (BESS) enclosures shall be performed according to the following cadence:

- Daily: Remote SCADA review of inverter fault codes, BESS state-of-charge, and thermal alarms.
- Monthly: On-site visual inspection of inverter enclosures, cooling systems, and BESS containers for signs of wear, corrosion, or thermal damage.
- Quarterly: Torque verification on DC and AC connections per manufacturer specification; thermal imaging scan of all electrical connections.
- Annually: Full functional test of protection relays, ground fault detection, and emergency shutdown systems.

4.2 Alarm Response Procedure

Upon receipt of a Priority 1 (critical) alarm from the SCADA system, operations personnel shall:

- Acknowledge the alarm within 5 minutes and assess whether remote remediation is possible.
- If remote remediation is not possible, dispatch a field technician within the contracted response window (see O&M Agreement, Section 7).
- Document the event, root cause, and corrective action in the asset management system within 24 hours of resolution.

4.3 Preventive Maintenance Constraints

Preventive maintenance activities requiring inverter or BESS de-energization must be scheduled during low-irradiance or low-dispatch windows where feasible, and require 48 hours advance notice to the grid operator when the maintenance affects available capacity greater than 5 MW.